AfDB · Data Innovation for Africa (DI4A)

NIGHT-TIME LIGHTS SERIES · THREE NOTEBOOKS

Night-time lights: from satellite to statistics

Collect, understand and analyse NASA and NOAA night-time light data to measure economic activity — one ISO3 code in, a public dashboard out

01 Collect

02 Explore and understand

03 Analyse and publish

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AGENDA

Six steps, from the night sky to a published indicator

01
The series
Why night lights measure activity, and how the three notebooks fit together

02
Collect
Notebook 1: find, test and download the right NASA Black Marble files

03
Explore and understand
Notebook 2: open the file, decode it, filter it, clip it to the country

04
Analyse
Notebook 3: trends, regions, GDP, shocks, inequality, industrial zones

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Publish and check
Static dashboard, reproducibility manifest, quality checks and limits

06
Key takeaways
What to verify before publishing any figure

SECTION

01

The series

Why can satellite images of the night say something about an economy?



Light seen from space is a trace of human activity

~463 m
pixel size (15 arc-seconds)

Every night
since 19 January 2012 (Black Marble)

01:30
local overpass time of the satellite

What lights capture well

Urban activity, industry running at night, electrification, power cuts, displacement, sudden shocks

What they see poorly

Agriculture, daytime trade, much of the services sector and the rural informal economy

How to use them

As a complement to official statistics: direction, timing and geography of activity — not a replacement for GDP

Sources: Román et al. (2018); Henderson, Storeygard & Weil (2012); Chen & Nordhaus (2011)

Three notebooks, one chain, one parameter

Notebook 1 · Collect

NASA Black Marble (VNP46A1–A4)

- Country extent and VIIRS tiles
- NASA CMR catalogue query
- Availability matrix, verified download

Out: .h5 files, inventory, session log

Notebook 2 · Explore

Anatomy of a Black Marble file

- HDF5 tree, layers, attributes
- Decoding, quality flags, masks
- Mosaic, clip, distribution

Out: GeoTIFF, CSV, metadata JSON

Notebook 3 · Analyse

VIIRS VNL V2 + Black Marble on Earth Engine

- Trends, regions, convergence
- GDP link, shocks, inequality
- Industrial zones

Out: dashboard on GitHub Pages

SECTION

02

Collect

How do we get the right NASA files, and only those?



One ISO3 code in, a verified collection out

1 Environment

2 Dependencies

3 Token (4 sources)

4 Parameters

5 Extent and tiles

6 CMR queries

7 Availability test

8 Verified download

9 Quality control

Steps 1–7: no account needed

Step 8: Earthdata token

Rerun safely (idempotent)

NASA Black Marble: four products, one grid

VNP46A1

Daily · raw

At-sensor radiance, no correction

Use

Diagnostics

VNP46A2

Daily · corrected

Lunar BRDF-corrected night

Use

Dated events: outages, disasters

VNP46A3

Monthly composite

Clouds and outliers removed

Use

Seasonality, monthly series

VNP46A4

Annual composite

The most stable of the four

Use

Trends, regional comparison

Grid 10° × 10° tiles | **2,400 × 2,400 pixels per tile** | **nW·cm^{-2}·sr^{-1}** | **HDF-EOS5 (.h5)** | **Collection 2 (v002)**

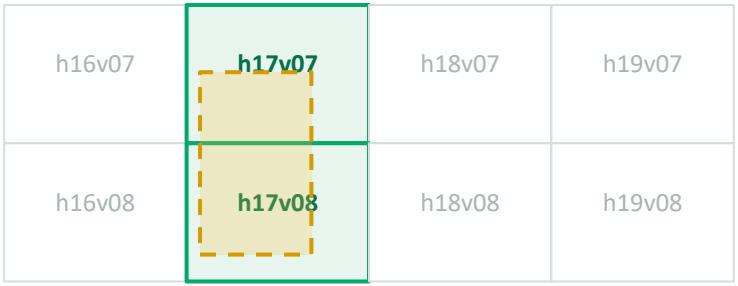
Sources: Román et al. (2018); NASA Black Marble User Guide, Collection 2.0 (2025); LAADS DAAC

From a country code to tiles, by simple arithmetic

Tile index

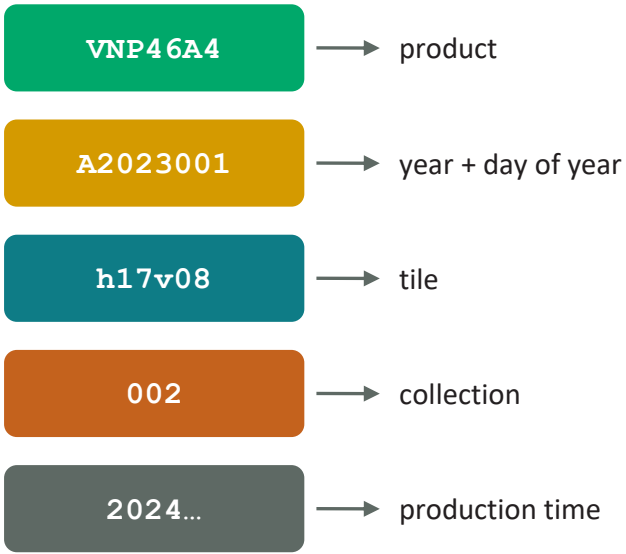
```
h = ⌊(longitude + 180) / 10⌋
v = ⌊(90 − latitude) / 10⌋
```

h: 0 → 35 west to east · v: 0 → 17 north to south

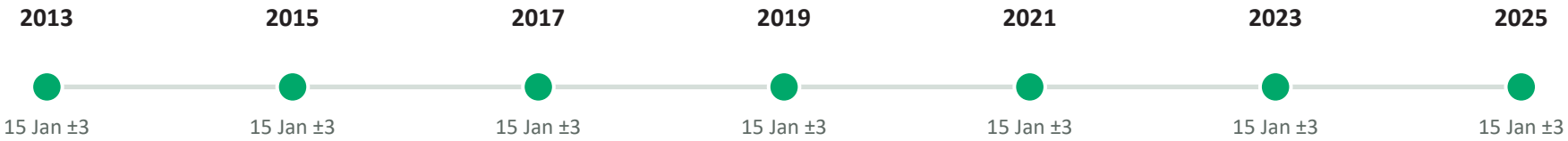


Côte d'Ivoire → h17v07 + h17v08

Anatomy of a file name



Fix the season, vary the year, never trust a single night



Daily A1 / A2

Fixed day ± 3 nights (7 nights per year), then keep clear, high-quality nights and take their median

Monthly A3

The same calendar month every year: the composite already absorbs clouds and the Moon

Annual A4

The whole year: the right product for trends across years

Moon: cycle of 29.5 days — a fixed date does not fix the lunar phase.

Ask the catalogue, measure availability, then download

1 · Query

One request per year · pagination (2,000 per page) · deduplication by tile and date

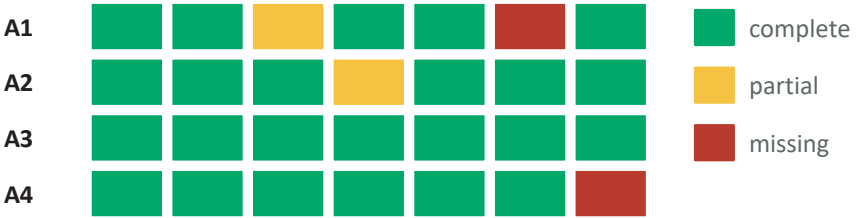
2 · Measure

Expected = windows × tiles × nights · found · completeness · volume in GB

3 · Download

Resume · size check · HDF5 opening · manifest

Coverage matrix (schematic)



Security by design

The token comes from an environment variable, a .env file, Colab or Kaggle secrets, or a masked prompt — and is sent only to *.earthdata.nasa.gov and *.eosdis.nasa.gov.

SECTION

03

Explore and understand

What is really inside a Black Marble file?



A Black Marble file is a binder, not an image

FILE VNP46A4 … .h5 — *global attributes: sensor, extent, production date*

HDFEOS / GRIDS / <grid> — *grid name depends on the product — never hard-code it*

Data Fields — *all useful layers live here*

Radiance
`AllAngle_Composite_Snow_Free`

Uncertainty
`…_Num` · `…_Std`

Quality
`…_Quality` · `Mandatory_Quality_Flag`

Context
`Land_Water_Mask` · `DNB_Platform`

Coordinates
`lat` · `lon`

Decoding: one equation and one trap

radiance = stored value × scale_factor + add_offset

…after setting every _FillValue pixel to NaN — attributes always read from the file

Daily layer (A2)

`uint16 · × 0.1 · fill 65535`

1247 → 124.7 nW

+6,553.5

per empty pixel if fill is ignored

Composite (A3 / A4, Collection 2)

`float32 · × 1 · fill −999.9`

124.7 → 124.7 nW

−999.9

per empty pixel if fill is ignored

Rule: the fill value is not a zero, and still less a measurement.

Which pixels can we trust? Quality flags and masks

Composite quality codes (A3 / A4)

0	Good quality · > 3 nights	kept
1	Poor quality · ≤ 3 nights	dropped
2	Gap-filled from history	dropped
255	Fill value	dropped

QA_MAX = "auto" reads this legend in the file

Land / water mask

Ocean and sea removed: moonglint, ships and fisheries are not black. Without it, coastal and landlocked countries are not comparable.

Cloud mask (daily)

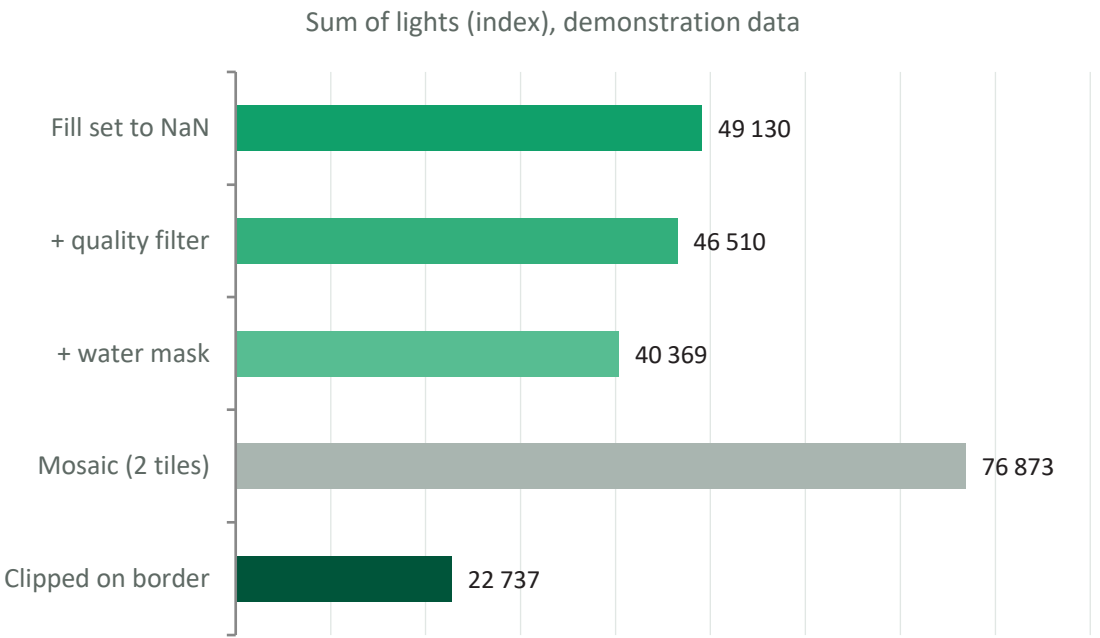
Bits 6–7: confident clear → confident cloudy. Value = (integer >> 6) & 3.

Number of nights

…_Num tells how many nights support each pixel: map it next to the radiance.

Source: NASA Black Marble User Guide, Collection 2.0 (2025)

The sum of lights, corrected step by step



No precaution

−70.7 M

fill value −999.9 summed as light

Final answer

22,737

national territory, quality-filtered, water masked

Notebook 2 demonstration set (simulated values, Côte d'Ivoire, Collection 2 structure)

Mosaic, clip, and respect a very skewed distribution

Mosaic

Same grid, same resolution: tiles pasted by index, no distortion

Clip

Point-in-polygon on the national border (official boundaries can replace the built-in ones)

Distribution

Log scale; median and sum say different things; check extreme pixels

Area weighting

Pixel area ∝ cos(latitude): −13 % at 30°, −20 % at 37°

Exports GeoTIFF · CSV · NPZ · metadata JSON · processing log

SECTION

04

Analyse

How do we turn radiance into economic indicators?



One pipeline, six analytical scenarios

02 Parameters → **03** Acquire & clean → **04** Explore → **05–10** 6 scenarios → **11** Key messages → **12–13** Export & publish

S1 · Time dynamics

Is activity rising, and how fast?

Seasonally adjusted series · YoY · trend %/yr · anomalies · breaks

S2 · Regional growth

Which regions grow, and do they converge?

Growth class · top-1 / top-10 % shares · HHI · β and σ convergence

S3 · Lights and GDP

Do lights track GDP, and can they fill gaps?

Elasticity β · nowcast · MAE vs naive · regional GDP

S4 · Shocks

How deep was the shock, and how fast the recovery?

Trough % · gap vs trend · months to recovery · placebo · daily index

S5 · Inequality & access

How unequal is light, and who lives in the dark?

Gini · Lorenz · Theil T · urban/rural ratio · population in the dark

S6 · Industrial zones

Where does the country produce?

Industrial score · zone share and trend · precision / recall vs OSM

Night-light data: four products, four roles

VIIRS VNL V2 · monthly

`NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG`

2014 → · ~464 m · avg_rad (nW·cm⁻²·sr⁻¹) + cf_cvg

Monthly mean radiance, stray-light corrected; cf_cvg = cloud-free nights

Role: S1 seasonality, S3 nowcast, S4 shocks

VIIRS VNL V2.1 / V2.2 · annual

`NOAA/VIIRS/DNB/ANNUAL_V21 · _V22`

2013–2021 / 2022 → · ~464 m · average_masked

Annual radiance with background and ephemeral lights masked by EOG

Role: Annual levels for S1–S3, S5, S6; background mask

Black Marble VNP46A2 · daily

`NASA/VIIRS/002/VNP46A2`

2012 → · ~500 m · BRDF / Moon corrected

Nightly radiance; quality flag 0–1, clear sky, no snow

Role: S4 daily index around events

DMSP-OLS · harmonised

`NOAA/DMSP-OLS · sat-io Harmonized_NTL`

1992–2013 (+ DMSP-like VIIRS) · ~1 km · DN 0–63

Digital numbers, saturate at 63 in city centres; DN > 7 kept

Role: S1 long series, S3 long-run elasticity

Sources: Elvidge et al. (2017, 2021); Román et al. (2018); Li et al. (2020) · IDs checked in the Earth Engine catalogue at each run

Who lives there, and what is built there?

GHSL GHS-POP R2023A `JRC/GHSL/P2023A/GHS_POP`	**100 m · 1975–2030, 5-year epochs · population_count**	Per-capita light (S2, S5), inhabited pixels ≥ 5 people, population in the dark, flare test
WorldPop `WorldPop/GP/100m/pop`	**~100 m · 2000–2020 · population**	Alternative population source (pop_source = "worldpop")
GHSL GHS-SMOD R2023A `JRC/GHSL/P2023A/GHS_SMOD_V2-0`	**1 km · 5-year epochs · smod_code (degree of urbanisation)**	Urban if code ≥ 21 → urban / rural split of light and of the population in the dark (S5)
GHSL GHS-BUILT-S R2023A `JRC/GHSL/P2023A/GHS_BUILT_S`	**100 m (from 10 m) · built_surface, built_surface_nres**	Non-residential share for the industrial score (S6); “not built-up” test for gas flares
ESA WorldCover v200 `ESA/WorldCover/v200`	**10 m · 2021 · 11 land-cover classes**	Optional: tree, crop, built, bare, water shares per unit — not used in the default run

Sources: Schiavina et al. (2023); Pesaresi et al. (2023), JRC GHSL R2023A; WorldPop (2018); Zanaga et al. (2022)

Boundaries, official statistics and validation data

Sentinel-5P TROPOMI NO_2

`COPERNICUS/S5P/OFFL/L3_NO2`

2018-07 → · ~1.1 km grid · tropospheric column (mol/m²)

Role: Annual mean per unit; Spearman ρ with light (S2); 20 % of the industrial score (S6)

geoBoundaries CGAZ · FAO GAUL

`WM/geoLab/geoBoundaries/600` · `FAO/GAUL/2025`

ADM0 country · ADM1 regions · ADM2 districts

Role: Units over which every pixel is summed; GAUL 2025, then 2015, if geoBoundaries fails

World Bank WDI (API)

`NY.GDP.MKTP.KD` · `EG.ELC.ACCS.ZS` · `SP.POP.TOTL` …

8 indicators · annual · national

Role: Real GDP (S3), electricity access total / urban / rural (S5), population check (±30 %)

OpenStreetMap (Overpass)

`landuse = industrial`

Volunteered polygons · completeness varies by country

Role: Validation only: precision and recall of candidate zones (S6)

Sources: Copernicus Sentinel-5P (ESA); Runfola et al. (2020); FAO GAUL 2025; World Bank WDI; © OpenStreetMap contributors (ODbL)

Which data feeds which scenario?

	Clean	S1	S2	S3	S4	S5	S6
VIIRS VNL monthly	●	●		●	●		
VIIRS VNL annual	●	●	●	●		●	●
Black Marble VNP46A2					●		
DMSP harmonised	○	●		●			
GHS-POP / WorldPop	●		●			●	●
GHS-SMOD						●	
GHS-BUILT-S	●						●
Sentinel-5P NO_2			○				●
geoBoundaries / GAUL	●	●	●	●	●	●	●
World Bank WDI	○			●		○	
OpenStreetMap							○

● input of the metrics ○ check, context or validation

From billions of pixels to two tables

Clean pixels

~464 m

↓

Sum per unit

ADM0 · ADM1 · ADM2

↓

Cached tables

+ parameter signature

Monthly table · unit × month

`sol`
sum of lights (observed + imputed)

`valid_frac`
share of the unit's pixels observed

`sol_adj`
sol if valid_frac ≥ 0.8, else missing

`imputed_share`
share of light imputed for clouds

Annual table · unit × year

`sol` · `sol_raw`
clean light · light before flare mask

`lit_km2` · `area_km2`
area ≥ 0.5 nW · total area

`pop` · `pop_inhabited`
population · in pixels ≥ 5 people

`pop_dark_<t>`
inhabited, radiance < threshold t

`pop_urban` · `sol_urban`
population and light where SMOD ≥ 21

`flare_sol`
light removed as gas flares

Remove everything that is not human activity

1 · Clouds

cf_cvg ≥ 2 · cloudy pixel = its annual value · unit-month kept if ≥ 80 % observed

2 · Negatives

Set to 0: calibration noise over dark areas

3 · Outliers

Cap 1000 nW · > 5 × annual median → median (fires, lightning, glints)

4 · Background

Unlit in annual composite → 0; newly lit pixels kept for unpublished years

5 · Gas flares

≥ 50 nW, < 5 inhabitants, not built-up → removed with a 2 km halo

6 · DMSP

Saturation (DN 63) measured; harmonised series thresholded at DN > 7

Source: Li et al. (2020) for the DMSP threshold · all thresholds are parameters of NTLConfig

Base metrics and quality checks, used everywhere

METRIC	DEFINITION / FORMULA	HOW TO READ IT
Sum of lights (SoL)	Σ radiance of clean pixels in the unit	An index of activity; compare across time, with care across latitudes
Light per capita	SoL / population	Active vs merely populated area
Lit area share	area with radiance ≥ 0.5 nW / unit area	Spatial extent of electrification and settlement
Flare share	flare_sol / sol_raw	High → oil and gas light dominates; results less reliable
Imputed share	light imputed for cloudy pixels / SoL	High → the month relies on annual values

- Population GHSL / World Bank in [0.7 ; 1.3]
- Jump at VNL V2.1 → V2.2 vs neighbouring years
- Jump at DMSP → VIIRS (2013 → 2014)

Time dynamics: is activity rising, and how fast?

QUESTION

Is light activity rising, falling or flat — and is a recent movement real or seasonal?

DATA USED

VIIRS VNL monthly
sol_adj per unit and month, 2014 →

VIIRS VNL annual
sol per unit and year

DMSP–VIIRS harmonised
1992 →, read as a trend only

METHOD, STEP BY STEP

1. **Regularise**
   Complete monthly calendar; gaps ≤ 3 months filled log-linearly
2. **Decompose (robust STL)**
   log(SoL) = trend + season (period 12) + remainder
3. **Seasonally adjust**
   SA = exp(trend + remainder): the curve to read
4. **Compare year on year**
   Each month vs the same month a year earlier
5. **Estimate the annual trend**
   Theil–Sen + Mann–Kendall on annual composites; regional (ADM1) series

Time dynamics: the metrics

METRIC	DEFINITION / FORMULA	HOW TO READ IT
Seasonally adjusted SoL	exp(trend + remainder)	The level net of the usual seasonal pattern
Year-on-year change	SoL_m / SoL_{m-12} − 1	Seasonality cancels out; noisy month to month
Annual growth (%/yr)	exp(Theil–Sen slope of log SoL) − 1, 95 % CI	Robust trend; read the interval, not only the point
Mann–Kendall p-value	Test of a monotonic trend	Confidence: high $p < 0.01$ · medium $p < 0.10$ · low otherwise or $n < 6$
Anomaly	$\lvert r - \text{med}(r)\rvert$ / (1.4826·MAD) ≥ 3.5 and $\lvert$deviation$\rvert$ ≥ 5 %	A month out of the ordinary, to explain
Structural break	Binary segmentation, linear segments ≥ 12 months, BIC-type penalty	Lasting change in level or slope

Sources: Cleveland et al. (1990) STL; Sen (1968); Mann (1945), Kendall (1975)

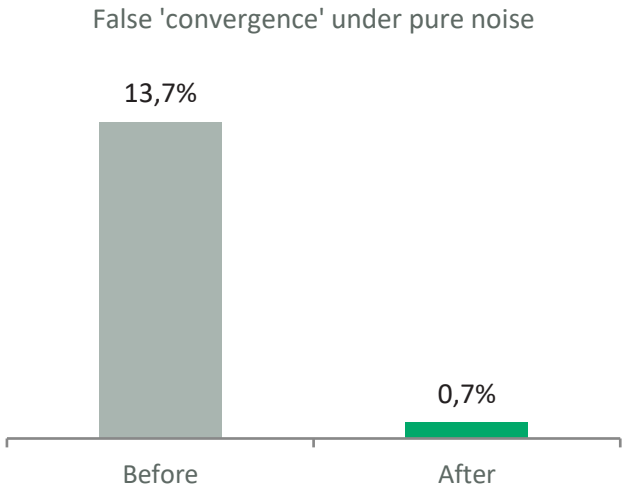
Regional growth: who grows, and do regions converge?

DATA USED

- VNL annual SoL per ADM1 / ADM2
- GHS-POP → light per capita
- Sentinel-5P NO_2, last year

METHOD, STEP BY STEP

1. **Robust growth per unit**
   Theil–Sen slope of log SoL, Mann–Kendall p
2. **Correct for multiple tests**
   Benjamini–Hochberg adjusted p (false discovery rate)
3. **Classify**
   Growth / Decline if |slope| ≥ 1 %/yr and p_adj < 0.10, else Stagnation
4. **Concentration and convergence**
   Top shares, HHI, β (3-year initial mean), σ; NO_2 correlation



Simulation: 60 units, 12 years, 25 % noise, no true convergence — share of runs wrongly significant at 95 %

Regional growth: the metrics

METRIC	DEFINITION / FORMULA	HOW TO READ IT
Growth class	$\|g\| \geq 1$ %/yr and BH-adjusted $p < 0.10$	Growth / Decline / Stagnation (= not established)
Top-1 and top-10 % shares	SoL of the brightest unit(s) / total SoL	Weight of the capital or main pole
Herfindahl index (HHI)	Σs_i^2 ($s_i = SoL_i / \Sigma SoL$)	0 → dispersed · 1 → all light in one unit
β-convergence	$g_i = a + \beta \cdot \log(\text{initial light/cap.}_i) + \varepsilon$, HC1 errors	$\beta < 0$ with CI < 0 → lagging units catch up
Half-life	$\ln 2 / -\ln(1 + \beta)$	Years needed to close half the gap
σ-convergence	st. dev. of log(light/cap.) across units, per year	Falling σ → regions become more alike
NO_2 correlation	Spearman ρ(NO_2 mean, SoL) across units	High ρ → light reflects combustion activity

Sources: Benjamini & Hochberg (1995); Barro & Sala-i-Martin (1992); Herfindahl (1950)

Lights and GDP: measuring the elasticity

QUESTION

Does light track official GDP, and by how much does it move when GDP moves by 1 %?

DATA USED

VIIRS VNL annual, national
SoL per year (+ V2.1 / V2.2 flag)

Harmonised long series
1992 →, sensor dummy

World Bank NY.GDP.MKTP.KD
Real GDP, constant 2015 USD

METHOD, STEP BY STEP

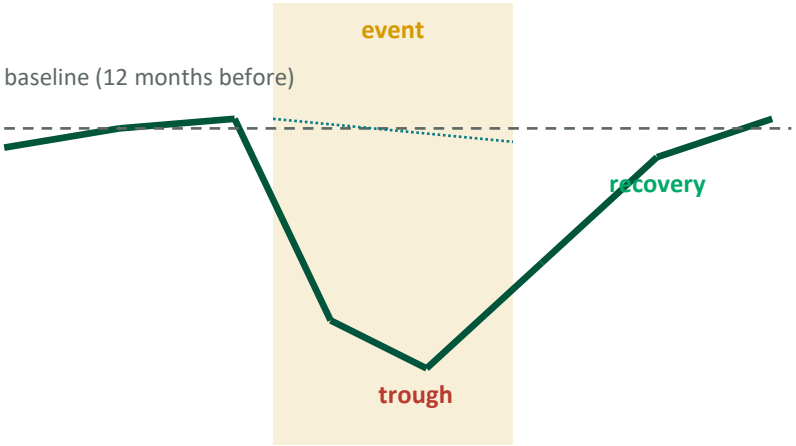
1. **Levels**
   $\log(SoL_t) = \alpha + \beta \cdot \log(GDP_t)\ [+ \delta \cdot V22_t] + \varepsilon_t$
2. **Growth rates**
   $\Delta\log(SoL) = \alpha + \beta \cdot \Delta\log(GDP)$: immune to common trends
3. **Long series**
   Levels on 1992 → with a DMSP / VIIRS dummy
4. **Uncertainty**
   Newey–West (HAC) 95 % CI + pairs bootstrap CI (999 draws)
5. **Diagnostics**
   Durbin–Watson · Breusch–Pagan · Jarque–Bera · Ramsey RESET ($n \geq 12$)

Lights and GDP: nowcast and regional estimate

METRIC	DEFINITION / FORMULA	HOW TO READ IT
Elasticity β	slope of log SoL on log GDP (or of growth rates)	+1 % GDP ↔ +β % light; Henderson's 0.3 ≈ 1/β
Confidence of β	high: n ≥ 15, p < 0.01, CI width < 1, R^2 ≥ 0.6	medium if p < 0.10 · low if n < 8 or not significant
Nowcast	Δlog GDP = a + γ·Δlog SoL, chained from last official year	Estimate for unpublished years, with 80 / 95 % bands
Pseudo-real-time MAE	mean |actual − predicted| growth, pp; expanding window ≥ 5 yrs	Typical error of a forecaster using lights
Relative MAE	MAE lights / min(MAE last-year, MAE past-mean)	< 1 → lights add information beyond naive rules
Regional GDP (experimental)	$GDP_i = GDP \cdot SoL_i^{\gamma} / \Sigma SoL_j^{\gamma}$, γ = 1 ; 0.5	Sums to national GDP; gap γ=1 vs 0.5 = uncertainty

Sources: Henderson, Storeygard & Weil (2012); Newey & West (1987) · regional GDP is labelled experimental everywhere

Shocks: how deep, how long, and is it real?



Schematic: seasonally adjusted series around an event

DATA USED

Seasonally adjusted monthly VNL (S1) · Black Marble VNP46A2 daily · event catalogue: COVID-19, CUSTOM_EVENTS, ACLED / EM-DAT

METRIC	FORMULA	READING
Trough %	100·(exp(min SA − baseline) − 1)	Depth of the fall
Gap vs trend %	mean deviation from the 24-month pre-trend	Loss vs where activity was heading
Months to recovery	first month SA ≥ baseline (≤ 36)	Speed of recovery; none = not recovered
Placebo test	trough < 5th percentile of earlier windows	Stands out from ordinary noise

Shocks: daily zoom and regional impact

Keep good nights

Quality flag ∈ {0, 1} · cloud bits 6–7 ≤ 1 · no snow · pixel lit in the annual composite

Daily sum

Day kept if ≥ 50 % of pixels observed; SoL scaled by the observed share

Smooth

7-day rolling median: absorbs lunar residuals and cloudy days

Index

100 × SoL / same days 364 days earlier (same weekday)

Reading the daily index

100	same activity as a year earlier
< 100	activity below last year (outage, lockdown, conflict)
> 100	activity above last year (recovery, electrification)

Regional impact and unexplained breaks

- Heat table: trough % by ADM1 × event (−30 % → +30 %).
- Breaks with no catalogued event within ±6 months → document before interpreting.

Inequality: how unevenly is light shared?

QUESTION

Is light per capita concentrated in a few places, and at which level should policy act?

DATA USED

VIIRS VNL annual
SoL per district (finest level)

GHS-POP
population weights, per-capita light

GHS-SMOD
urban if code ≥ 21

METHOD, STEP BY STEP

1. **Light per capita by district**
   $x_i = SoL_i / pop_i$, weight $w_i = pop_i$
2. **Gini and Lorenz curve**
   Cumulative population share vs cumulative light share
3. **Theil T and its decomposition**
   T = between regions (ADM1) + within regions
4. **Change and its uncertainty**
   ΔGini first → last year, bootstrap 300 paired draws, 95 % CI
5. **Urban / rural gap**
   Light per capita urban / rural

Population in the dark: a proxy for access to electricity

METRIC	DEFINITION / FORMULA	HOW TO READ IT
Gini	weighted; 0 = equality, 1 = all light in one unit	Overall concentration of light per capita
Theil T	$\Sigma\, w_i\, (x_i/\mu)\, \ln(x_i/\mu)$, w_i population shares	Splits into between- and within-region parts
Urban / rural ratio	light/cap. where SMOD ≥ 21 ÷ elsewhere	How much more light an urban resident gets
Population in the dark	Σ pop of pixels with ≥ 5 people and radiance < t	Published for t = 0.25 · 0.5 · 1 nW: a range
Dark share	pop in the dark / inhabited population	Proxy of the share without electricity
Gap with surveys (pp)	(1 − dark share) − WDI access rate EG.ELC.ACCS.ZS	Large gap → low-consumption access or data issue

Sources: Theil (1967); Schiavina et al. (2023) GHSL; World Bank WDI · flare halos excluded from the count

Industrial zones: four signals, one score per pixel

SIGNAL	DATA	NORMALISATION (0–1)	WEIGHT
Non-residential built-up	GHSL BUILT-S	nres share / 0.6	**35 %**
Light intensity	VNL, median last 3 yrs	$\log_{10}$(radiance) / 2	**30 %**
NO_2 emissions	Sentinel-5P, annual mean	(NO_2 − med) / (p98 − med)	**20 %**
Few residents	GHSL GHS-POP	1 − $\log_{10}(1 + \text{hab/km}^2) / \log_{10}(5000)$	**15 %**



Industrial score

$$S = 0.35 \cdot \text{built_nres} + 0.30 \cdot \text{light} + 0.20 \cdot NO_2 + 0.15 \cdot \text{few_residents} \quad \text{(each signal clipped to 0–1)}$$

Weights and thresholds: expert choices in IND.PARAMS, to be tested · Sources: Pesaresi et al. (2023); Copernicus Sentinel-5P

Industrial zones: from pixels to validated candidates

METRIC	DEFINITION / FORMULA	HOW TO READ IT
Candidate pixel	radiance ≥ 5 nW, built ≥ 1 %, S ≥ 0.55 — or gas flare	Neighbours (≥ 2 pixels) → zone; max 250 zones
Share of national light	SoL of zone / national SoL (last year)	Economic weight of the zone
Zone dynamics	Theil–Sen growth: > +5 %/yr expanding, < −3 % declining	Opening, expansion or slowdown signal
Class	energy: flare share > 0.3 & nres < 0.3 · major: top 10 % SoL or ≥ 8 km²	Energy site · major hub · industrial / logistics zone
Precision	zones intersecting an OSM industrial area / zones	≥ 0.5 → medium confidence, else low
Recall	OSM industrial areas found / OSM areas	Coverage; depends on OSM completeness
Light-weighted precision	SoL of confirmed zones / SoL of all zones	Is the bulk of industrial light confirmed?

From metrics to key messages

High — Established result: can be quoted with its interval

Medium — Plausible result: quote with caution and context

Low — A lead, not a conclusion: investigate before using

Messages are generated in EN and FR and fill the notebook's executive summary (section 00)

WHERE THE CONFIDENCE COMES FROM

Trend (S1, S2)

high $p < 0.01$ · medium $p < 0.10$ · low if $n < 6$

Elasticity (S3)

n, p, CI width, R^2 (see S3)

Shock (S4)

placebo test passed or not

Inequality (S5)

bootstrap CI of ΔGini excludes 0

Industry (S6)

OSM precision ≥ 0.5 → medium, else low

SECTION

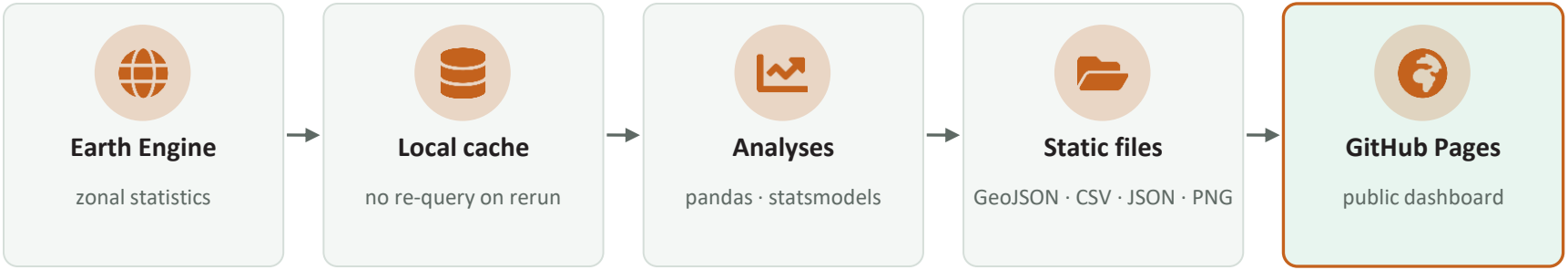
05

Publish and check

How do results reach users, and how far can they be trusted?



A static, bilingual dashboard published in one call



Bilingual EN / FR

Same figures, texts generated in each language

Reproducibility manifest

Parameters, versions, dataset availability

Secure publication

Fine-grained token, 7–30 days, never written anywhere

Key messages with confidence: high · medium · low

Synthetic test country 'XTS': public publishing refused

Checked before it ships

0

execution errors in notebooks 1 and 2, EN and FR

13.7%→0.7%

false β-convergence under pure noise (simulation)

29 → 0

spurious growth classes out of 300 null districts

- Attributes and fill values read from each file
- Parameters and code version invalidate a stale cache
- Every export travels with its metadata and log
- Quality legend and water codes read, not assumed
- GHSL population checked against the World Bank (±30 %)
- Simulated values always flagged as such

Six artefacts to know before concluding anything

Blooming

Light spills beyond built-up areas

Saturation

Brightest cores plateau; DMSP saturates at 63

Gas flares

Bright without local economic activity

Seasonality

Clouds and fires create non-economic cycles

Sensor breaks

DMSP → VIIRS, VNL V2.1 → V2.2, collections

Rural noise

Weakest signal where the rural poor live

Sources: Gibson et al. (2021); Elvidge et al. (2021); Li et al. (2020)

SECTION

06

Key takeaways

What should a statistician remember?



Twelve checks before publishing a night-lights figure

1. File names intact; year read from the name
2. Data group searched, not assumed
3. Scale, offset and fill read from the file
4. Fill values set to NaN first
5. North at the top: row 0 is the northernmost
6. Quality threshold documented
7. Ocean and sea masked
8. Number of nights inspected
9. Brightest pixel plausible
10. Same layer and version across the series
11. Right product for the question
12. Limitations written down

Trend on A4, season on A3, event on A2



Choose the product

Trend → annual A4 · season → monthly A3 · dated event → daily A2. Never the other way round.



Respect the order

Fill → NaN, quality, water, mosaic, clip. Each skipped step changes the figure silently.



Complement, not substitute

Direction, timing and geography of activity — always with official statistics and documented limits.

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One ISO3 code in, a public dashboard out

Collect · Explore and understand · Analyse and publish

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